

Data Engineer Technical Assessment

Overview

Assume you are working for a financial services company that provides **digital wallet** and **card issuing** services.

The digital wallet allows customers to perform several types of financial activities, including:

- Top up their wallet through a merchant or a linked bank account.
- Send money to, and receive money from, other customers using the service.
- Purchase goods within the application, either through instant payment or three-month installments.
- Top up a card to make purchases on international e-commerce platforms.
- Cash out through ATMs or authorized agents.

Each customer may have one or more accounts.

The Entity Relationship Diagram is provided in the following file:

erd_diagram.jpg

The source data is provided in the following file:

data_tables.zip

You are required to use the provided ERD and data files to complete the tasks below.

Task 1: Operational Database

Install an operational database product of your choice on your local machine.

Using the provided ERD, create the required database tables and load the provided data into the database.

You should submit any scripts or files used for:

- Creating the database.

- Creating the tables.
- Loading or ingesting the data.

You may choose any operational database product that you think is suitable for this scenario.

Task 2: Data Warehouse and ETL

Install a data warehouse or analytical database solution of your choice.

Create the required structures in the data warehouse and develop ETL or ELT scripts to move the data from the operational database into the data warehouse.

You should submit any scripts or files used for:

- Creating the data warehouse tables.
 - Extracting data from the operational database.
 - Transforming the data, if needed.
 - Loading the data into the warehouse.
 - Updating the data in the data warehouse.
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Submission Requirements

You may submit your solution in one of the following ways:

1. GitHub Repository

- Provide a link to a GitHub repository that contains all scripts, code, and documentation.

2. Attached Files

- Submit the scripts and code files directly as part of your assessment submission.

Regardless of the submission method, you must include a document that clearly describes:

- The operational database product used.
- The data warehouse or analytical database product used.
- Any assumptions made during the implementation.
- Any observations or issues found in the provided data.
- If using GitHub, make sure the repository is accessible and organized clearly enough for review.